# MAPS OF SIMPLICIAL SYSTEMS LIFT TO MORPHISMS OF CARTESIAN OBJECTS UNDER A VANISHING ASSUMPTION

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the parenthetical '(omitted)' (source line 3122) covers the check that X_1 -> X_0 -> K' vanishes: X_1 -> X_0 is the alternating sum of the transitions and composing with the adjoint of K_0 -> K'_0 yields the alternating sum of the system maps, which vanishes by the definition of a morphism of simplicial systems - an unpacking of the adjunction; the closing 'this easily gives' (source line 3170) is the determination of the components gamma_n from gamma_0 through the transition isomorphisms of the cartesian object K, which forces {gamma_n} to equal the given system morphism; both are auxiliary to the fully written induction.

*Editorial note.* The label lemma-hom-cartesian-objects-derived names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- The Postnikov system associated to a cartesian object. Same chapter, \label{lemma-abelian-postnikov}. Lines 3115-3118: the objects X_n and Postnikov system Y_n attached to K; Y_0 = X_0 and the triangles Y_1 -> X_1 -> Y_0 -> Y_1[1].
- Cartesian objects identified by the shriek-tensor maps. Same chapter, \label{lemma-derived-cartesian-shriek} (card L46169 of this batch). Lines 3118-3120: cartesianness of K gives X_n = g_{n!}K_n.
- Adjunction between g_{n!} and restriction to components. Same chapter, \label{lemma-restriction-to-components-site} (card L46149). Lines 3133-3137: Hom(X_{n+1}[n], K') = Hom(K_{n+1}[n], K'_{n+1}).
- Morphisms of simplicial systems of the derived category. Same chapter, the definition of morphisms of simplicial systems. Lines 3115-3117 and 3120-3126: compatibility of the maps {K_n -> K'_n} with the system structure underlies the vanishing of X_1 -> X_0 -> K'.

*Source reference.* Original reference 0D9I: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3002–3002. Prerequisite pointer invoked at proof line(s) 3119; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0D9E: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2752–2752. Prerequisite pointer invoked at proof line(s) 3120; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 3104; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *In Situation 09WE. If $K, K' \in D(\mathcal{C}_{total})$. Assume*

(1) *$K$ is cartesian,*
(2) *$\mathrm{Hom}(K_i[i-1], K'_i) = 0$ for $i > 1$.*

*Then any map $\{K_n \to K'_n\}$ between the associated simplicial systems of $K$ and $K'$ comes from a map $K \to K'$ in $D(\mathcal{C}_{total})$.*

*Proof.* Let $\{K_n \to K'_n\}_{n \geq 0}$ be a morphism of simplicial systems of the derived category. Consider the objects $X_n$ and Postnikov system $Y_n$ associated to $K$ of Lemma 0D9I. By (1) and Lemma 0D9E we have $X_n = g_{n!}K_n$. In particular, the map $K_0 \to K'_0$ induces a morphism $X_0 \to K'$. Since $\{K_n \to K'_n\}$ is a morphism of systems, a computation (omitted) shows that the composition

$$X_1 \to X_0 \to K'$$

is zero. As $Y_0 = X_0$ and as $Y_1$ fits into a distinguished triangle

$$Y_1 \to X_1 \to Y_0 \to Y_1[1]$$

we conclude that there exists a morphism $Y_1[1] \to K'$ whose composition with $X_0 = Y_0 \to Y_1[1]$ is the morphism $X_0 \to K'$ given above. Suppose given a map $Y_n[n] \to K'$ for $n \geq 1$. From the distinguished triangle

$$X_{n+1}[n] \to Y_n[n] \to Y_{n+1}[n+1] \to X_{n+1}[n+1]$$

we get an exact sequence

$$\mathrm{Hom}(Y_{n+1}[n+1], K') \to \mathrm{Hom}(Y_n[n], K') \to \mathrm{Hom}(X_{n+1}[n], K')$$

As $X_{n+1}[n] = g_{n+1!}K_{n+1}[n]$ the last group is equal to

$$\mathrm{Hom}(K_{n+1}[n], K'_{n+1})$$

which is zero by assumption (2). By induction we get a system of maps $Y_n[n] \to K'$ compatible with transition maps and reducing to the given map on $Y_0$. This produces a map

$$\gamma : K = \mathrm{hocolim} Y_n[n] \longrightarrow K'$$

This map in any case has the property that the diagram

$$\begin{array}{ccc} X_0 & \longrightarrow & K \\ & \searrow & \downarrow{\scriptstyle \gamma} \\ & & K' \end{array}$$

is commutative. Restricting to $\mathcal{C}_0$ we deduce that the map $\gamma_0 : K_0 \to K'_0$ is the same as the first map $K_0 \to K'_0$ of the morphism of simplicial systems. Since $K$ is cartesian, this easily gives that $\{\gamma_n\}$ is the map of simplicial systems we started out with. $\square$